

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Internal Hackathon

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5- Internal Hackathon	Assessment:	Assessment Tool 3 – Internal Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
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Section / Batch:	SLOT-D	Date:	29-08-2026

Code of Conduct

I Abzam A / 192573029 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

PROBLEM STATEMENT:

Diabetes/Heart Disease Risk Prediction. Binary classification of disease presence using clinical parameters

(BMI, glucose, blood pressure, age) with the Pima Indians Diabetes dataset or similar; evaluate with

confusion

matrix,

ROC-AUC.

OUTPUT SCREENSHOTS:

The screenshot shows a web application interface for entering clinical parameters. At the top, there's a title 'Clinical Parameter Entry' and a subtitle 'Enter patient biometric measurements for binary risk classification.' To the right, there are 'Quick Profiles' buttons: 'Sample 1: Low Risk' (Normal Range), 'Sample 2: Prediabetes' (Moderate), and 'Sample 3: High Risk' (High Risk). Below this is a 'Patient Record ID / Reference (Optional):' field with a placeholder 'e.g. PT-2026-0042'. The main form consists of four sliders: 'Blood Glucose Concentration' (50 to 250 mg/dL, current value 110, range 'Prediabetes (100-125)'), 'Diastolic Blood Pressure' (40 to 140 mm Hg, current value 78, range 'Optimal (< 80 mm Hg)'), 'Body Mass Index (BMI)' (15.0 to 55.0 kg/m², current value 26.5, range 'Overweight (25.0-29.9)'), and 'Patient Age' (18 to 90 years, current value 42, range 'Standard Demographic (< 45)'). At the bottom left is a 'Reset Fields' button, and at the bottom right is a 'Calculate Risk & Save Log' button.

⚠️

Evaluation Report • Record #20

Critical Risk of Diabetes

Copy Text

Print Report

ESTIMATED RISK PROBABILITY

94%

statistical score

0% Safe

Threshold 50%

100% High

RISK STRATIFICATION

Binary: 1

CRITICAL RISK

Biometric profile presents markers consistent with diabetes diagnosis.

TELEMETRY RECORD

Model:

Logistic Regression

Timestamp:

4:00:00 AM

Database Status:

Logged in SQLite

CLINICAL BIOMETRICS EVALUATION

Blood Glucose

178 mg/dL

Diabetic Range

Glucose exceeds clinical threshold (≥ 126 mg/dL), representing significant glycemic stress.

Body Mass Index (BMI)

37.2 kg/m²

Obese

BMI indicates clinical obesity (≥ 30.0 kg/m²), strongly associated with cellular insulin insensitivity.

Diastolic Blood Pressure

94 mm Hg

Hypertension Stage 1/2

Diastolic BP is clinically hypertensive (≥ 90 mm Hg), increasing cardiovascular and metabolic risk.

Age Factor

56 years

Higher Demographic Risk

Age (≥ 50) is an established epidemiological risk multiplier for metabolic resistance.

CLINICAL GUIDANCE & RECOMMENDED NEXT STEPS

Schedule a comprehensive clinical screening including HbA1c test and Oral Glucose Tolerance Test (OGTT).

Consult with an endocrinologist or primary care physician for diagnostic confirmation.

Adopt a low-glycemic dietary pattern rich in dietary fiber, reducing refined carbohydrates and added sugars.

Implement moderate structured physical activity (e.g. 150 min/week) and caloric balance to target a 5-7% weight reduction.

Monitor sodium intake (<2,300 mg/day) and follow the DASH diet protocol to optimize arterial blood pressure.

Stay well hydrated and ensure 7-8 hours of consistent restorative sleep to regulate cortisol and insulin sensitivity.

Note: This statistical model is an adjunct tool for clinical decision-support and does not replace standardized laboratory tests (HbA1c / FPG) or medical consultation.

Confusion Matrix Table

Validation outcomes on 58 unseen test observations

TRUE NEGATIVE (TN)

22

37.9% of total

Correct non-diabetic

FALSE POSITIVE (FP)

14

24.1% of total

Type I Error

FALSE NEGATIVE (FN)

8

13.8% of total

Type II Error

TRUE POSITIVE (TP)

14

24.1% of total

Correct diabetic

The model prioritizes sensitivity to minimize missed diabetic profiles while retaining standard diagnostic specificity.

ROC Curve (Receiver Operating Characteristic)

AUC: 0.7311

True Positive Rate (%) vs False Positive Rate (%)

False Positive Rate (%)	True Positive Rate (%)
0	0
5	10
10	20
15	30
20	40
25	50
30	60
35	65
40	70
45	75
50	80
55	85
60	90
65	95
70	98
75	99
80	100
100	100

Model Curve (AUC 0.7311) Diagonal Chance (AUC 0.50)

Clinical Feature Importance Weights

Relative algorithmic weight of each clinical indicator in the classification model

Glucose

43.2%

Primary predictor of glycemic regulation

BMI

33.7%

Adiposity biomarker for insulin resistance

Age

22.9%

Demographic physiological risk multiplier

BloodPressure

0.3%

Vascular endothelial biomarker

TOTAL LOGGED

20

Stored in SQLite

POSITIVE / HIGH RISK

50%

10 cases

COHORT AVG GLUCOSE

146.95 mg/dL

Clinical Average

COHORT AVG BMI

37.35 kg/m²

Clinical Average

Patient Assessment Audit Log

Historical record of patient evaluations persisted in SQLite database.

Q Search records...

All Categories

Export CSV

RECORD ID	PATIENT TAG	GLUCOSE	BLOOD PRESSURE	BMI	AGE	BINARY OUTCOME	RISK SCORE	CATEGORY	ACTIONS
#20	PT-003	178 mg/dL	94 mm Hg	37.2 kg/m²	56 yrs	1 (Positive)	94.0%	Critical Risk	
#19	UNSPECIFIED	110 mg/dL	78 mm Hg	26.5 kg/m²	42 yrs	0 (Negative)	33.5%	Moderate Risk	
#18	PT-003	250 mg/dL	140 mm Hg	55 kg/m²	90 yrs	1 (Positive)	100.0%	Critical Risk	
#17	PT-003	250 mg/dL	140 mm Hg	55 kg/m²	83 yrs	1 (Positive)	99.9%	Critical Risk	
#16	PT-003	250 mg/dL	140 mm Hg	55 kg/m²	49 yrs	1 (Positive)	99.8%	Critical Risk	
#15	PT-003	250 mg/dL	140 mm Hg	55 kg/m²	18 yrs	1 (Positive)	99.3%	Critical Risk	
#14	PT-003	50 mg/dL	40 mm Hg	55 kg/m²	18 yrs	0 (Negative)	33.7%	Moderate Risk	
#13	PT-003	50 mg/dL	40 mm Hg	15 kg/m²	18 yrs	0 (Negative)	1.2%	Low Risk	
#12	PT-003	250 mg/dL	140 mm Hg	55 kg/m²	90 yrs	1 (Positive)	100.0%	Critical Risk	
#11	UNSPECIFIED	110 mg/dL	78 mm Hg	26.5 kg/m²	42 yrs	0 (Negative)	33.5%	Moderate Risk	

Clinical Guidelines & Diagnostic Reference Standards

Diagnostic reference thresholds established by the American Diabetes Association (ADA) and WHO.

1. BLOOD GLUCOSE (MG/DL)

Plasma glucose concentration measures circulating blood sugar. It represents the primary direct metabolic indicator for diabetes screening.

Normal Fasting: < 100 mg/dL (70-99)

Impaired (Prediabetes): 100 - 125 mg/dL

Diagnostic Threshold: ≥ 126 mg/dL

2. DIASTOLIC BLOOD PRESSURE (MM HG)

Measures baseline vascular resistance between myocardial beats. Arterial hypertension frequently accompanies metabolic insulin resistance.

Normal Diastolic: < 80 mm Hg

Elevated / Prehypertension: 80 - 89 mm Hg

Hypertension Stage 1/2: ≥ 90 mm Hg

3. BODY MASS INDEX - BMI (KG/M²)

BMI calculates weight normalized for height squared. Adiposity is a strong biomarker associated with cellular insulin insensitivity.

Normal Weight: 18.5 - 24.9 kg/m²

Overweight: 25.0 - 29.9 kg/m²

Clinical Obesity: ≥ 30.0 kg/m²

4. CHRONOLOGICAL AGE (YEARS)

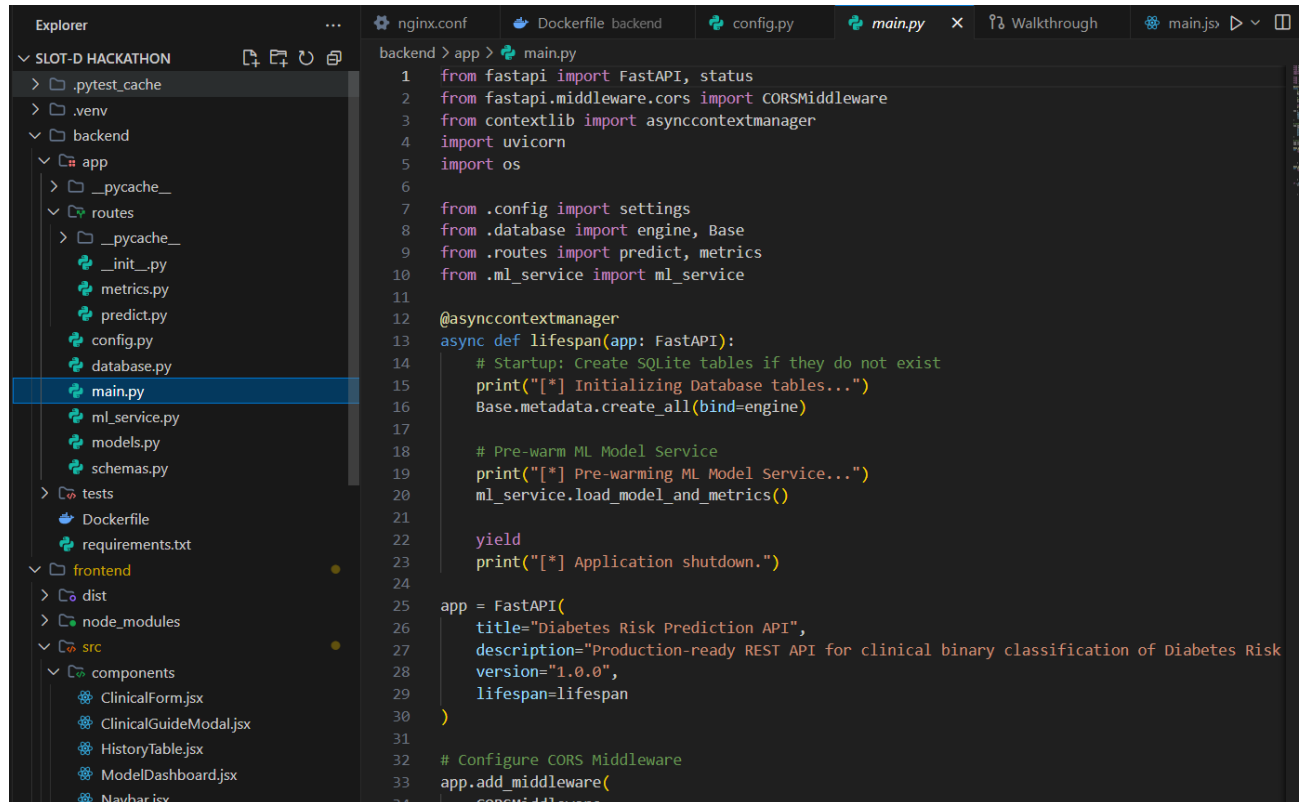
Age is an epidemiological risk factor for metabolic decline and adult-onset Type 2 diabetes.

Standard Baseline: < 35 years

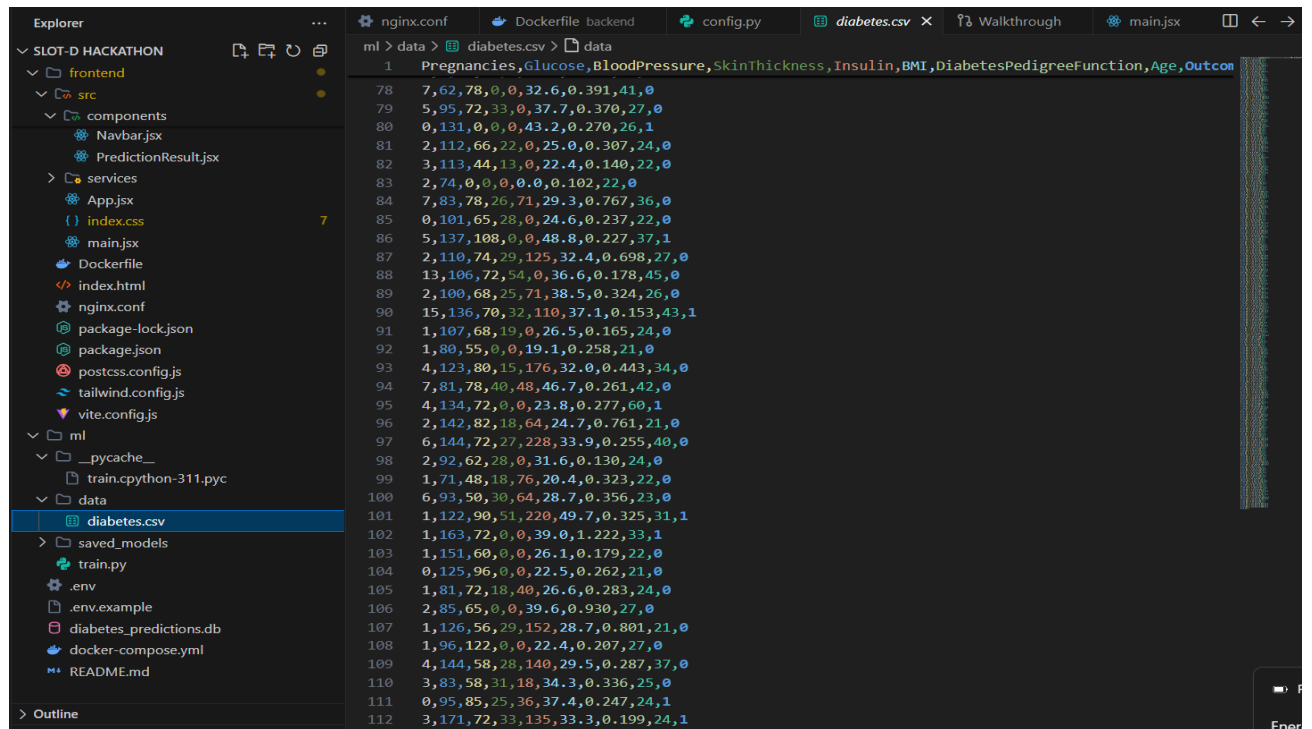
Moderate Age Risk: 35 - 44 years

Elevated Demographic Risk: ≥ 45 years

CODE :



```
1 from fastapi import FastAPI, status
2 from fastapi.middleware.cors import CORSMiddleware
3 from contextlib import asynccontextmanager
4 import uvicorn
5 import os
6
7 from .config import settings
8 from .database import engine, Base
9 from .routes import predict, metrics
10 from .ml_service import ml_service
11
12 @asynccontextmanager
13 async def lifespan(app: FastAPI):
14     # Startup: Create SQLite tables if they do not exist
15     print("[*] Initializing Database tables...")
16     Base.metadata.create_all(bind=engine)
17
18     # Pre-warm ML Model Service
19     print("[*] Pre-warming ML Model Service...")
20     ml_service.load_model_and_metrics()
21
22     yield
23     print("[*] Application shutdown.")
24
25 app = FastAPI(
26     title="Diabetes Risk Prediction API",
27     description="Production-ready REST API for clinical binary classification of Diabetes Risk",
28     version="1.0.0",
29     lifespan=lifespan
30 )
31
32 # Configure CORS Middleware
33 app.add_middleware(
34     CORSMiddleware
```



```
1 Pregnancies,Glucose,BloodPressure,SkinThickness,Insulin,BMI,DiabetesPedigreeFunction,Age,Outcomes
78 7,62,78,0,0,32.6,0.391,41,0
79 5,95,72,33,0,37.7,0.370,27,0
80 0,131,0,0,0,43.2,0.270,26,1
81 2,112,66,22,0,25.0,0.307,24,0
82 3,113,44,13,0,22.4,0.140,22,0
83 2,74,0,0,0,0,0.102,22,0
84 7,83,78,26,71,29.3,0.767,36,0
85 0,101,65,28,0,24.6,0.237,22,0
86 5,137,108,0,0,48.8,0.227,37,1
87 2,110,74,29,125,32.4,0.698,27,0
88 13,106,72,54,0,36.6,0.178,45,0
89 2,100,68,25,71,38.5,0.324,26,0
90 15,136,70,32,110,37.1,0.153,43,1
91 1,107,68,19,0,26.5,0.165,24,0
92 1,80,55,0,0,19.1,0.258,21,0
93 4,123,80,15,176,32.0,0.443,34,0
94 7,81,78,40,48,46.7,0.261,42,0
95 4,134,72,0,0,23.8,0.277,60,1
96 2,142,82,18,64,24.7,0.761,21,0
97 6,144,72,27,228,33.9,0.255,40,0
98 2,92,62,28,0,31.6,0.130,24,0
99 1,71,48,18,76,20.4,0.323,22,0
100 6,93,50,30,64,28.7,0.356,23,0
101 1,122,90,51,220,49.7,0.325,31,1
102 1,163,72,0,0,39.0,1.222,33,1
103 1,151,60,0,0,26.1,0.179,22,0
104 0,125,96,0,0,22.5,0.262,21,0
105 1,81,72,18,40,26.6,0.283,24,0
106 2,85,65,0,0,39.6,0.930,27,0
107 1,126,56,29,152,28.7,0.801,21,0
108 1,96,122,0,0,22.4,0.207,27,0
109 4,144,58,28,140,29.5,0.287,37,0
110 3,83,58,31,18,34.3,0.336,25,0
111 0,95,85,25,36,37.4,0.247,24,1
112 3,171,72,33,135,33.3,0.199,24,1
```

TEAMWORK:

